

CONTINUATION

Continue — Project Status Snapshot

Revision: 29 **Last modified:** 2026-08-27T18:20:00Z

COMMIT-RECORD CORRECTION — read this before trusting `git log`

Commit 5c9b9e0 carries the message **test-probe-do-not-use [skip-ci]**. That message is wrong and it is my error, not a description of the change. I invoked `scripts/commit-push-all.sh` with a throwaway string while diagnosing why an earlier invocation had exited 1, expecting a probe; the script's contract is "commit with this message", so it did exactly that — staged all 147 files, committed, and pushed to all three remotes.

Nothing was lost and nothing needs redoing. 5c9b9e0 contains the complete session batch; 80bd720 is the wrapper's automatic workable-items DB differential dump for it. The record is corrected forward here rather than by amend or force-push, which §11.4.113 forbids without exception once a commit is public on a mirror.

What 5c9b9e0 actually contains:

Area	Change
<code>scripts/ownership_repair.sh:994</code>	BOB-207 — the repair filter selected on BOTH uid and gid, so a file already owned by the operator but carrying a foreign gid was skipped by the pass meant to fix it. Now selects on uid only; the chown -h "\$

Area	Change
	<code>{OP_UID}:\${OP_GID}"</code> at :719 already wrote both, so this is one predicate, not a behaviour change. BOB-221 — an expected-RED declaration a suite must carry before its RED counts as evidence. 19/19, deterministic hash 80aecdb63bf348be. BOB-227 — these were UNTRACKED: present on disk, invisible to a fresh clone (the §11.4.233(G) absent-checkout class). Now tracked. PASS=197 FAIL=0. BOB-195 gate hardening (101 → 144 assertions over review rounds 12-15, gate script byte-identical throughout) merged onto 10 incoming upstream commits, pushed to all 8 constitution upstreams before the pointer bump. Evidence custody snapshots with MANIFESTs (§11.4.83 / §11.4.226). Rounds 9-11 worked on uncommitted files with no snapshot, so their round-over-round diff claims stay permanently uncheckable; this commit ends that.
scripts/lib/expected_red.sh + invariant 30	
scripts/pre_build/lan_route_auth_analyzer.py + gate + test	
constitution 7a5e53a → a09b1ea	
docs/qa/BOB-{102,186,191,195,196,198,205,206,207,212}	

Still owed, deliberately deferred, not forgotten: the long pre-build gate was skipped via BOBA_SYNC_SKIP_CI=1 and stamped [skip-ci] into the commit — the §11.4.234(D) recorded-deferral path, chosen because T042 is a known-open failure and the commit/push mechanism must stay unblocked. Clear it with:

```
bash scripts/commit-push-all.sh "catch-up long gate"
```

HOW TO RESUME THIS WORK

Point a fresh session at this file, then run the SpecKit execute command for feature 002-user-owned-downloads. Everything below is the live state at the moment of this write; every number in it was measured, not remembered.

```
/speckit-superspec-execute 002-user-owned-downloads
```

ONE-PARAGRAPH STATE

Feature 002-user-owned-downloads is at **40 of 42 tasks complete**. The two open tasks are **T041** (final independent review of git diff main...HEAD) and **T042** (run the full 52-invariant gate with no skip flag). Neither can close yet: T042 will report **3 failures of 39** bash suites, all three of them correctly-authored RED tests for open defects, and four independent review chains are mid-iteration under §11.4.134 (iterate to zero findings AND zero warnings). All work is committed and pushed; nothing is in flight in a working tree.

THE FOUR REVIEW CHAINS — exact round state

Each chain is a code artifact plus its own adversarial review loop. §11.4.134 requires iterating to a zero-finding, zero-warning GO. None has reached one.

Chain	Artifact	Last completed	Verdict	Next action
BOB-102	LAN-route auth gate	round 13 remediation	—	round 14 review OWED (killed by session limit)
BOB-195		round 15 remediation	—	round 16 review

Chain	Artifact	Last completed	Verdict	Next action
	dangerous-combination gate			OWED (killed by session limit)
BOB-221	expected-RED mechanism	round 4 review	NO-GO 0B/1I/4M/3N	round 5 remediation
BOB-207	ownership uid/gid primitive	evidence correction	0 BLOCKING	short re-review

BOB-102 — LAN-route authentication gate

- **PASS=197 FAIL=0**, exit 0. Real tree: 74 routes / 3 services / 23 known-gaps, --json md5 ecd4b31c2f40c87ed321a00c43cacb26.
- Round 12 found the **real primitive had been misidentified for three rounds**: it is not “a selector can split at a dot” but *a construct whose ABSENCE disarms a refusal, rather than dropping a credit*. Silence is safe for a credit-carrier and unsafe for a refusal-armer. That sentence is now in-source.
- Round 13 fixed it at the class level and swept nine refusal-arming constructs across three axes, finding two more open doors (an aliased gin import, and all nine unmodelled-construct members) — both gofmt-stable.
- **Round 14 review never ran**. Its brief is reconstructable from this file plus the round-13 report; the highest-value task is verifying the “nine refusal-arming constructs” enumeration is complete, and adjudicating round 13’s correction of round 12 (it says the .Group(“collateral” was not new to the poison).

BOB-195 — dangerous-combination fail-closed gate

- **144 assertions, 0 failures, exit 0**. The gate script has been **byte-identical since round 11** (md5 b90c5945d44b891c793c448e7165682c) — every defect found in rounds 12-15 was under-pinning, never a live bug.
- Round 14 found the first door in the **primary AST analyzer** (all prior 13 rounds were in the degraded text fallback): excluding str constants from the trivial-literal predicate makes return “unknown” and return “” pass clean. Root cause measured: grep -c ‘return ‘ across the harness was **0**.
- Round 15 swept every AST predicate (22 mutations) and found **12 more doors**, worst being suppress(builtins.Exception) — the flagship violation with a qualified name — classified NARROW and dropped with no trace.

- **Round 16 review never ran.** Its first task is attacking three survivors round 15 classified as non-doors, and sweeping the two areas round 15 names as unswept: the bash driver and the non-Python scanners.

BOB-221 — expected-RED declaration mechanism (blocks T042)

- Suite **19/19**, deterministic x3 (80aecdb63bf348be). Empty-table parity proven byte-identical against HEAD.
- Round 4 review verdict: **NO-GO, 0 BLOCKING / 1 IMPORTANT / 4 MINOR / 3 NIT**. No path yields HONOURED for an undeclared suite — that was hunted specifically and not found.
- **IMPORTANT-1**: the conflicting-markers refusal is unvalidated instrumentation. Deleting it survives 19/19 and fails OPEN — a suite carrying markers for two different items is honoured when the table id sorts first. Needs a P19 scenario.
- **MINOR-1**: boundary row 7 claims duplicated markers BLOCK; measurement shows two byte-identical markers are deduped by sort -u and **HONOURED**. Doc-vs-code drift on a documented refusal; also the candidate twentieth boundary row.
- MINOR-2 the status cache is dead code at its only call site (subshell); MINOR-3 boundary 19 asserted by nothing; MINOR-4 the marker-MISMATCH reason is unpinned.

BOB-207 — ownership uid/gid primitive (T042 blocker, CLOSED)

- **Closed and verified.** test_ownership_gid_agreement.sh 5/0/0, test_ownership_repair.sh 146/0/0, test_check_cm_ownership_invariants.sh 42/42.
- The fix is one line: the repair's walk now selects on uid alone, while the chown still writes uid:gid. Design artifacts settle the direction — data-model.md E4 models the probe with probe_uid/expected_uid and no gid field anywhere.
- The suite was re-based off a gid proxy onto a **real foreign uid** via podman unshare chown 1:1 (host view 100000:100000), correcting a header that claimed such a fixture was impossible unprivileged. All 23 fixtures re-based, none dropped.
- Review returned 0 BLOCKING; one evidence-accuracy correction was applied.

T042 — WHY IT WILL FAIL, PRECISELY

Invariant 30 discovers 41 bash suites, excludes 2 self-recursive, runs **39**. It will report **3 failures**, all intentional REDs for open defects:

```
tests/unit/test_ownership_gid_agreement.sh      BOB-207  -> now
rc=0, PASSES
tests/pre_build/test_bob221_expected_red_declaration.sh  BOB-221
-> now rc=0
tests/pre_build/test_bob205_danger_roots_scope.sh  BOB-205  ->
still rc=1
```

Corrections to earlier analysis, measured: **there is no rc=124 and no rc=127**. Two suites *do* hit rc=124, but the driver excludes them by basename before the run loop, so T042 never sees them. Reading the raw glob instead of the loop is what produced the phantom.

Remaining T042 blockers: BOB-205 (still RED), plus a non-quiescent tree at gate time (invariant 30's no-trace half fails if exports move mid-run).

FILED THIS SESSION

```
BOB-226 Task major      repair-side foreign-owned interior
directories untested
BOB-227 Bug  critical   LAN-route auth gate artifacts were
UNTRACKED (closed by this commit)
BOB-228 Task minor      README does not link the gate guide
(11.4.212 orphan)
```

Tracker verified in sync: 227 items, DB and Markdown agree, all invariants satisfied.

Not re-minted, correctly deduped: the directory-setgid finding is already **BOB-220**; the orphan security test is already **BOB-212**.

FOUR LESSONS THIS SESSION PAID FOR

1. **Carrier-vs-thing bit five times.** A count matched its own summary line (three separate chains); a test oracle matched the word abort in its own scenario *filename*; a residue scan flagged prose describing the finding. BOB-221 fixed the class rather than the instance: every reason alternative must now be multi-word and every scenario path space-free, so a filename carrier is impossible by construction rather than by luck.

2. **Evidence custody failed in three chains independently.** Rounds worked on uncommitted files with no snapshot, so round-over-round diff claims became permanently uncheckable. BOB-195 and BOB-102 now snapshot reviewed bytes to docs/qa/<item>/round-N/ with a manifest before editing. **This commit makes those durable for the first time.**
3. **Baseline before discrepancy.** git diff against HEAD spans every round, not the one under review. Three times a wrong baseline nearly became a fabricated finding. Establish which baseline a number used before calling it a contradiction.
4. **The reviewer-authored mutation channel found the load-bearing defect in eight consecutive rounds.** A reviewer must write at least one mutation the author did not (11.4.194(6)(d)). An unmeasured redundancy dismissal should be inadmissible — round 8 of BOB-102 dismissed a live PASS-bluff by reading regexes without running one.

OPERATIONAL HAZARDS OBSERVED

- **The agent scratchpad is shared.** One stream overwrote another's mutation helper mid-run, producing 8 false negatives while the anchor was present throughout. Namespace scratch artifacts per-agent and re-verify any surprising negative before reporting it.
- **Backgrounded runs killed by a tool ceiling never fire their EXIT trap** — 32 GB of orphaned fixture dirs accumulated in /tmp before a sweep. An age-gated sweep-on-start is the compensation.
- **A mode=ro SQLite open LEAVES -wal/-shm sidecars behind; a read-write open removes them on clean close.** Sidecar presence therefore proves nothing about write access — the opposite of the intuitive reading.